

2D Shapes & Attributes

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **polygon** is a closed shape made of straight sides. We name polygons by their number of sides and angles: a **triangle** has 3 sides and 3 angles, a **quadrilateral** has 4, a **pentagon** has 5, and a **hexagon** has 6. The **quadrilateral family** includes the **square** (4 equal sides, 4 right angles), the **rectangle** (4 right angles, opposite sides equal), the **rhombus** (4 equal sides), and the **trapezoid** (at least one pair of parallel sides).

Round 1 — Retrieve It

1. How many sides and angles does a triangle have?

2. What is the name for a polygon with 4 sides?

3. How many sides does a hexagon have?

4. Name two shapes that belong to the quadrilateral family.

Round 2 — Sketch It

Draw four boxes labeled triangle, quadrilateral, pentagon, and hexagon. Inside each box, draw the polygon and write how many sides it has.

2D Shapes & Attributes — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A shape has 5 straight sides and 5 angles. What polygon is it, and how do you know?

2. A quadrilateral has 4 equal sides and 4 right angles. What is it called, and why?

3. How is a rectangle different from a square? Name one thing that is the same and one that is different.

Round 4 — Reason from Evidence

A shape has 4 sides, but only one pair of sides is parallel and its sides are not all equal. A friend says it must be a square. Use what you know about the quadrilateral family to explain why the friend is wrong.

2D Shapes & Attributes — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how you can name a polygon just by counting. Use the words sides and angles.

C**Connect**

Connect these shapes to something you have seen, such as a stop sign, a floor tile, or a window. Name the polygon and its number of sides.

E**Extend**

A square, a rectangle, and a rhombus are all quadrilaterals. Describe what makes each one special within the quadrilateral family.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — 2D Shapes & Attributes

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. 3 sides and 3 angles.
2. A quadrilateral.
3. 6 sides.
4. Any two of: square, rectangle, rhombus, trapezoid.

Sketch It — look for

Triangle drawn with 3 sides, quadrilateral with 4 sides, pentagon with 5 sides, hexagon with 6 sides. Side counts should match each label.

Apply It

1. A pentagon; a polygon with 5 sides and 5 angles is called a pentagon.
2. A square; a quadrilateral with 4 equal sides and 4 right angles is a square.
3. Same: both have 4 right angles and 4 sides. Different: a square has all 4 sides equal, while a rectangle needs only its opposite sides equal.

Reason from Evidence — look for

A square must have 4 equal sides and 4 right angles. This shape has only one pair of parallel sides and unequal sides, so it is a trapezoid, not a square.

ACE — Extend look-for

Square: 4 equal sides and 4 right angles. Rectangle: 4 right angles with opposite sides equal. Rhombus: 4 equal sides (angles need not be right angles).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.